DRUMPANEL · PRODUCT REQUIREMENTS AND SPECIFICATIONS

DrumPanel product requirements and specifications

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.6	2026-10-24	Amish Chadha	Kit 1.8 upgrade: requirements migrated from DMP-REQ-001 to product requirements PR-001 to PR-012 (old R number in Was) with tolerance, units, the user requirement each traces to (DMP-URS-001) and the TRL of verification; PR-013 to PR-016 added from the constraints of DMP-PRB-001 (drum acceptance, purge water, hand power only, pictogram procedure); no target changed. STEM fixes F1 to F5 of 2026-10-24 (DMP-DEC-001): per-drum flatness check (PR-005), second drain stand (PR-007 now 2.20), LEL channel calibration (PR-002), slot tolerance (PR-010), two-person head lifts (PR-011), fire-resistant hydraulic fluids refused (PR-013); cost USD 3,566 (PR-009)
0.5	2026-10-03	Amish Chadha	Round-2 decision 5A (DMP-DDR-004): R7 kept at 2.5 drums an hour; the TRL

			4 timed trial decides; no design change
0.4	2026-10-03	Amish Chadha	R7 restated to 2.5 drums an hour by Amish on 2026-10-03 (decision 5A, DMP-DDR-003); status of R5, R6, R7, R9 and R11 from DMP-CAL-001 v0.2 after the notching punch, ring head and dial adjuster
0.3	2026-10-03	Amish Chadha	TRL 3; status of every requirement from DMP-CAL-001 on the constructable design (DMP-DDR-002); R9 worded against the value-engineering target; R12 changed to three panels per drum
0.2	2026-10-03	Amish Chadha	TRL 2; targets firmed up; R10 guarding, R11 handling and R12 yield added; concept status for each requirement
0.1	2026-09-30	Amish Chadha	Initial scaffold

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What the bench set must do, each requirement traced to a user need in DMP-URS-001 (links in DMP-RTM-001). The requirements describe option A, the current design (DMP-OPT-001; the concept choice is proposed, awaiting Amish). On paper the constructable design meets 14 of the 16 requirements, misses one and is over the value-engineering target of another: PR-007 (throughput) is **not met**, about 2.20 drums an hour against 2.5, and PR-009 (cost) is over the USD 3,000 target by USD 566. PR-001 to PR-012 are the requirements R1 to R12 of DMP-REQ-001 v0.5 with the same targets, verification and status; PR-013 to PR-016 write down four constraints of DMP-PRB-001 as requirements. On 2026-10-03 Amish chose option A on every requirement decision put to him ("1A 2A 3A 4A 5A 6A 7A 8A 9A 10A 11A"; DMP-DDR-003) and kept R7 at 2.5 drums an hour in round 2 ("i agree with all the 46 recommendations you provided. please proceed."; DMP-DDR-004). Every figure is from DMP-CAL-001 v0.3 (`docs/04-calcs/sizing.py`); the tag in brackets is the line of its output. Targets remain proposals for review with a co-design partner. Method: STANDARDS section 22.

Table 1. Product requirements and specifications. Status from DMP-CAL-001 v0.3 (2026-10-24).

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	Was
PR-001	UR-001	Safety	No open flame	No step from drum to flat sheet shall use flame or heat; no grinder or torch touches a drum	0	none	heat or flame steps	Inspection of the procedure and build plan, TRL 3; observed trial, TRL 4	Met by design: purge, check, cut and roll are all cold	R1
PR-002	UR-002	Safety	Vapour check before cutting	The vapour check shall flag any drum with flammable vapour at or above 10 % of the lower explosive limit (LEL), at both bungs, before any cut, with the detector's LEL channel calibrated on pentane or set with the maker's correction factor for heavy hydrocarbon vapour	10	alarm set at 5	% LEL	Bump test of the bought detector with 50 % LEL gas, TRL 4; bench test with a calibrated gas, TRL 4	Met by design: bought pumped LEL detector, alarm at 5 % LEL, half the target (DMP-DDR-001, D2)	R2
PR-003	UR-003	Performance	Remove ends cold	One person shall take both heads off a 208 L drum in under 10 minutes	10	max	min	Analysis (CAL section 8), TRL 3; timed trials on a batch of drums, TRL 4	Met on paper: about 3.9 min [H7]; crank force 12 to 27 N [C10 to C15]	R3
PR-004	UR-004	Performance	Open the side	The slit shall run straight along the full body length	5	max deviation	mm	Inspection of the model (rail straight within 1 mm), TRL 3; measure	Met by design: the shear head runs on a rail	R4

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	Was
								trial drums, TRL 4		
PR-005	UR-005	Performance	Flat sheet	Panels shall be flat within 10 mm over 1 m after rolling and hand-setting the ends	10	max over 1 m	mm	Analysis (CAL section 5), TRL 3; straightedge and feeler check, TRL 4	Met on paper, with a trial pass for each batch and a straightedge check of the first panel of every drum (fix F1): the bending roll needs 0.29 mm setting accuracy [E13], 11.8 divisions of the 0.025 mm dial [E14d], and 0.02 mm of wall or 5 MPa of yield uses that up, so the dial is trimmed drum by drum and a panel out of tolerance is passed again; panel ends set by hand [E15, E16]	R5
PR-006	UR-006	Usability	Manageable effort	Every crank and lever force in normal use shall be at or below 150 N at the handle	150	max	N	Analysis (CAL sections 3 to 5, 9), TRL 3; spring scale on each handle, TRL 4	Met on paper: end cutter 27 N, shear head 68 N (slit) and 136 N (two ring cuts) at 1.5 mm, notching punch 131 N, slip roll	R6

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	Was
									20 N [C15, D15, D15r, K3, E23]	
PR-007	UR-007	Performance	Throughput	A two-person team shall turn at least 2.5 drums an hour into flat sheet (restated by Amish on 2026-10-03, was 4)	2.5	min	drums/h	Analysis (CAL section 8), TRL 3; timed half-day workshop trial, TRL 4	**Not met** on paper: about 2.20 drums an hour [H5, H13], 54.6 hands-on minutes a drum against 48.0 [H4, H9], with two drain stands so the purge keeps up [H12]; kept at 2.5 by Amish (DMP-DDR-004); the TRL 4 trial decides	R7
PR-008	UR-008	Physical architecture	Local build	Every part shall be made by welding, drilling and turning in a town with a welder and a lathe, or bought as a commodity part	All parts	pass or fail	n/a	Inspection of the drawings, BOM and build plan, TRL 3; build by a partner fabricator, TRL 4	Met by design: the rolls need a lathe with 1.1 m between centres; discs, cutter wheel, bearings, gears and chain are bought	R8
PR-009	UR-009	Cost	Value engineering	Bench set parts shall cost at or below the USD 3,000 value-engineering target (a hypothetical control	3,000	max	USD	Priced BOM (`bom/bom.csv`), TRL 3	Over the target: estimated USD 3,566, USD 566 over [J1 to J3]	R9

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	Was
				target, not a spending limit)						
PR-010	UR-010	Safety	Guarding	No finger shall reach a roll nip, disc nip, gear or chain in normal use; guard slots shall be made 7 to 8 mm wide; guards shall need a tool to remove	ISO 13857 Table 4	slots 7 to 8 mm	n/a	Analysis (CAL section 7), TRL 3; guards as built checked by a competent person, TRL 4	Met on paper: every slot 8 mm or less and 45 mm or more from its nip [G1 to G3]; chains in closed cases	R10
PR-011	UR-011	Safety	Handling	No lift over 25 kg by one person; the shear head and ring head are lifted to the rail by two people; a drum full of water shall never be lifted	25	max one person	kg	Model masses (CAL section 10), TRL 3; weigh parts, TRL 4	Met with a two-person rule: heaviest piece the 33.7 kg cradle frame [I4]; shear head 23.7 kg, ring head 20.4 kg, punch 18.7 kg [I5, I7, I9]; a full drum (about 240 kg) is filled and drained in place [B2]	R11
PR-012	UR-012	Performance	Yield	Each drum shall give three flat panels of about 233 x 1,800 mm and two head discs about 550 mm across	233 x 1,800; 550	±5; ±5	mm	Analysis (CAL section 1), TRL 3; measure trial output, TRL 4	Met by design: panels 233, 234 and 233 x 1,800 mm; heads 552 mm [A1 to A3]	R12
PR-013	UR-015	Safety	Drum acceptance	Only drums whose label	60	min flash point	°C	Inspection of the	Met by design:	new

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	Was
				shows a non-volatile oil (lubricating, mineral hydraulic or vegetable oil, flash point above 60 °C) shall be accepted; fire-resistant hydraulic fluids (phosphate esters, water glycols, polyalkylene glycols), which do not separate from water, and all others are returned				acceptance procedure sheet, TRL 3; observed intake, TRL 4	acceptance is step 1 of the procedure and the first pictogram sheet (BOM line 22)	
PR-014	UR-013	Environment and durability	Purge water	Residue and purge water shall drain into a tray, settle in a spare drum and be reused; skimmed oil shall go to a closed container for a recycler	0	none to ground	L	Inspection of the purge kit and procedure, TRL 3; observed purge, TRL 4	Met by design: drain tray, settling drum with tap and hose kit (BOM lines 2 and 3)	new
PR-015	UR-006	Usability	Hand power only	Every station shall be worked by hand cranks, levers and screws; no electricity, fuel or compressed air	0	none	powered drives	Inspection of the model and BOM, TRL 3	Met by design	new
PR-016	UR-014	Usability			6	min	sheets			new

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	Was
			Pictogram procedure	Acceptance, purge, vapour check, cutting, rolling and finishing shall each have a laminated pictogram sheet				Inspection, TRL 3; review with apprentices in their language, TRL 4	Met by design: BOM line 22; content drafted from DMP-BLD-001	

Categories: physical architecture, performance, materials, usability, packaging and transport, safety, environment and durability, cost. Status: met on paper, met by design, at risk, not met, over target, withdrawn, to verify at TRL 4.

Requirement restated on 2026-10-03

R7 (now PR-007) was "at least 4 drums to flat sheet per hour for a two-person team". Amish restated it to 2.5 drums an hour when he chose option 5A on 2026-10-03 ("1A 2A 3A 4A 5A 6A 7A 8A 9A 10A 11A"): add a lever notching punch and a second shear head so two ring cuts are made per pass, and restate R7 to 2.5 drums an hour. The record is DMP-DDR-003.

In round 2 Amish kept R7 at 2.5 drums an hour (5A of DMP-DDR-004): "i agree with all the 46 recommendations you provided. please proceed." The paper estimate (2.26 drums an hour then, 2.20 since the STEM fixes of 2026-10-24) was not changed by that decision, and PR-007 is not claimed as met. The gap is about 12 % of the hands-on time, inside the uncertainty of the task-time estimates, so the TRL 4 timed half-day trial decides, and no design change is made at TRL 3.

Assumptions

- Most drums reaching artisans held oils or similar contents that a water purge can clear enough to cut cold. Drums that did not are rejected at the door (PR-013).
- Artists accept painted panels, with paint removal as a separate cold step (DMP-DDR-001, D4).
- Drum body steel has a yield strength of 200 to 300 MPa and a wall of 0.8 to 1.2 mm; heads are 1.0 to 1.5 mm.
- The rolling hoops are cut out as two strips, so the slip roll never has to straighten them (DMP-DDR-002, P7).
- A pumped LEL detector, bump-tested each day, is good enough for a pass or fail check before cold cutting.
- The 150 N force limit of PR-006 is a conservative screen for a two-hand crank in a standing posture; an EN 1005-3 assessment belongs to TRL 4.

Safety

Safety: PR-001, PR-002, PR-010, PR-011 and PR-013 are safety requirements. They are not traded against cost or throughput: where a cheaper or faster option would weaken them, the conservative option was taken (DMP-DEC-001). The bench set is an open engineering reference, never certified equipment.